

BUSA8000 – Techniques in Business Analytics

Session 2, 2024

ASSESSMENT TASK	Assessment 2
STUDENT NAME	Chayut Deekongsieng
STUDENT ID	46800883
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Use of resources and technologies including generative artificial intelligence

Acknowledgement Statement by students:

- ☒ I acknowledge that I have **only used** GAITs (e.g., ChatGPT) in drafting and proofreading this assignment, which is permitted in the assignment instructions.

Section 1: Exploratory Data Analysis

Basic Data information

	koala_id	region	habitat	gender	age_in_years	Paw Size	total_length	head_length	ear_size	foot_length	skull_width	eye_diameter	chest_circumference	belly_circumference
0	1	1	VIC	m	8.0	16.0	77.0	19.1	19.5	24.5	15.4	2.2	63.0	61.0
1	2	1	VIC	f	6.0	16.5	79.5	17.5	16.2	22.5	12.6	3.0	63.5	58.0
2	3	1	VIC	f	6.0	19.0	83.5	19.0	16.9	25.4	15.0	2.5	65.0	59.0
3	4	1	VIC	f	6.0	18.0	80.0	18.2	17.2	26.1	12.1	2.2	63.0	59.0
4	5	1	VIC	f	2.0	16.0	73.5	16.5	18.2	21.0	11.3	2.1	63.5	58.0

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 104 entries, 0 to 103
Data columns (total 14 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   koala_id             104 non-null    int64
1   region               104 non-null    int64
2   habitat              104 non-null    object
3   gender               104 non-null    object
4   age_in_years         102 non-null    float64
5   Paw Size             104 non-null    float64
6   total_length         104 non-null    float64
7   head_length          104 non-null    float64
8   ear_size             104 non-null    float64
9   foot_length          102 non-null    float64
10  skull_width           103 non-null    float64
11  eye_diameter          104 non-null    float64
12  chest_circumference  104 non-null    float64
13  belly_circumference  103 non-null    float64
dtypes: float64(10), int64(2), object(2)
memory usage: 11.5+ KB
```

```
print("Shape of Koala Dataset", koala.shape)
Shape of Koala Dataset (104, 14)
```

First, the 'koala' dataset was loaded into a pandas DataFrame. This dataset provides detailed information about individual koalas, focusing on their physical traits and characteristics. It contains 104 rows and 14 columns, with both categorical and numerical features such as paw size, head length, and foot length. Metadata like region, habitat, and gender are also included. Upon reviewing the dataset, some columns, including age_in_years, foot_length, skull_width and belly_circumference, were found to have missing values that need to be addressed. Additionally, the habitat and gender columns may have inconsistent or misspelled entries, which will be corrected during data cleaning.

Checking and Handling for Missing Values

After checking for null values using isnull(), 6 missing entries were identified in the dataset, represented as 'NaN'. The missing values were found in the age_in_years, foot_length, and belly_circumference columns, all of which are numerical. To handle these missing values appropriately, they should be replaced with suitable estimates. In this case, the mean value of each column was used. This method ensures that the missing data is filled with the average measurement, helping to maintain the overall distribution of the dataset. Before applying the mean, it was confirmed that it is not extreme, ensuring it is a reliable choice for filling in the missing values without significantly impacting the dataset's integrity.

	koala_id	region	habitat	gender	age_in_years	Paw Size	total_length	head_length	ear_size	foot_length	skull_width	eye_diameter	chest_circumference	belly_circumference
40	41	2	VIC	f	5.0	16.5	71.0	13.4	5.3	NaN	12.0	2.9	62.0	55.5
43	44	2	VIC	m	NaN	15.5	64.0	10.1	17.6	20.3	6.5	1.4	58.0	52.0
45	46	2	VIC	m	NaN	15.0	72.0	16.4	16.2	22.8	9.4	1.4	59.5	60.0
64	65	5	QLD	f	5.0	18.0	76.5	18.5	6.7	NaN	12.4	1.0	64.0	63.5
75	76	6	QLD	m	5.0	21.0	77.0	17.4	11.4	14.5	11.8	4.8	61.0	NaN
88	89	7	QLD	m	6.0	15.0	72.5	22.7	11.2	14.4	NaN	1.4	64.0	55.5

```
Mean of age_in_years:  3.8333333333333335
Mean of foot_length:   18.461764705882352
Mean of skull_width:   12.91747572815534
Mean of belly_circumference:  57.58252427184466
```

Correcting inconsistent and misspelled categorical entries

There are only two categorical columns in the dataset: habitat and gender. The unique() function was used to identify the values in each column, revealing inconsistencies in state names and gender entries, as

shown in the figure. To address this, corrections were applied by standardizing the habitat column to use consistent 3-character state codes (QLD, VIC) and formatting the gender column as either 'male' or 'female'.

Before Corrections

```
print(koala['habitat'].unique())
print(koala['gender'].unique())

['VIC' 'VICToria' 'Vic' 'QLD' 'Queensland' 'Queenstown' 'qld' 'q']
['m' 'f' 'female' 'male']
```

After Corrections

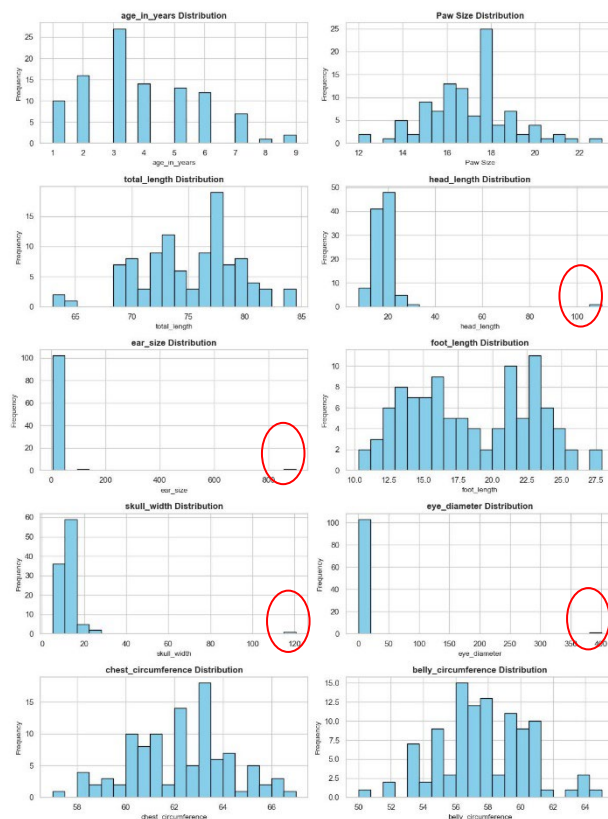
```
print(koala['habitat'].unique())
print(koala['gender'].unique())

['VIC' 'QLD']
['male' 'female']
```

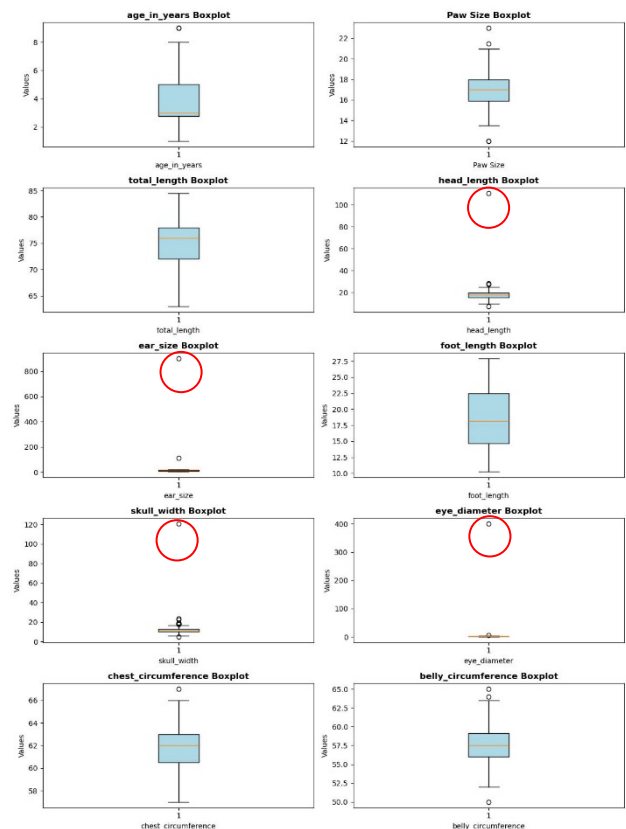
■ Checking the distribution of data, identifying and handling outliers.

Histograms were created for each numerical feature to examine the distribution, and box plot analysis was used to visualize and detect any extreme values that deviated significantly from the rest of the data.

Histogram of Koala Dataset Feature



Boxplots of Koala Dataset Feature



From the graphs, the following extreme outliers were identified: **ear_size = 900.8**, **eye_diameter = 400.5**, **head_length = 110.5**, and **skull_width = 120.6**. Upon reviewing these values, they were determined to be unrealistic and might require to be removed from the dataset.

However, to precisely identify outliers, the IQR (Interquartile Range) method was applied to define the lower and upper bounds of the data. Any values exceeding these bounds were treated as outliers. Additionally, the skewness of each column was calculated to assist in justifying the handling of outliers, providing insight into the distribution and potential impact of extreme values.

```
def detect_outliers(df, column):
    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    # Identifying outliers
    outliers = df[(df[column] < lower_bound) | (df[column] > upper_bound)]
```

Outlier Calculation

Skewness – before handling outlier

```
Skewness of age_in_years: 0.5442630373490612
Skewness of Paw Size: 0.16786870303351742
Skewness of total_length: -0.285265771534212
Skewness of head_length: 8.233721013431943
Skewness of ear_size: 9.992993956629151
Skewness of foot_length: 0.0907054084380688
Skewness of skull_width: 9.021706278249088
Skewness of eye_diameter: 10.188535641454523
Skewness of chest_circumference: -0.05551075216583214
Skewness of belly_circumference: 0.07841220995652716
```

>1

Outlier Results

Outliers in age_in_years:		Outliers in skull_width:	
age_in_years	skull_width	skull_width	
8	9.0	14	120.6
10	9.0	16	22.7
Outliers in Paw Size:		47	18.2
Paw Size		53	18.0
41	12.0	54	18.2
42	12.0	57	19.2
53	23.0	58	17.8
86	21.5	78	5.0
Outliers in head_length:		97	23.6
head_length		Outliers in eye_diameter:	
54	28.1	eye_diameter	
58	27.5	53	400.5
72	7.5	75	4.8
77	110.5	Outliers in chest_circumference:	
Outliers in ear_size:		chest_circumference	
ear_size		58	67.0
57	110.4	Outliers in belly_circumference:	
89	900.8	belly_circumference	
		20	65.0
		38	50.0
		56	64.0

Based on the obtained skewness values, several columns exhibit highly skewed distributions (skewness < -1 or skewness > 1), necessitating manipulation. These include head_length, ear_size, skull_width, and eye_diameter. In contrast, columns with moderately skewed distributions (-1 < skewness < -0.5 or 0.5 < skewness < 1), such as age_in_years, are less concerning. Other columns have skewness within the acceptable range (-0.5 < skewness < 0.5) and are not considered problematic.

After reviewing the outliers in relation to skewness, two approaches were taken:

- **Removing unrealistic extreme values:** This was applied to values that are clearly outliers compared to the rest of the data, including ear_size = 900.8 and 110.4, eye_diameter = 400.5, head_length = 110.5, and skull_width = 120.6.
- **Retaining plausible extreme values:** Values that, although extreme, appear reasonable within the context of the dataset were kept. This included outliers such as age_in_years = 9, as well as outliers in skull_width and belly_circumference, where the values did not differ dramatically from the rest.

```
# Remove rows contained unrealistic extreme values
koala = koala[~koala['ear_size'].isin([900.8, 110.4])]
koala = koala[koala['eye_diameter'] != 400.5]
koala = koala[koala['head_length'] != 110.5]
koala = koala[koala['skull_width'] != 120.6]

koala.shape

(99, 14)
```

Skewness – after 1st handling outlier

```
Skewness of age_in_years: 0.5623976380148998
Skewness of Paw Size: -0.014512497232162318
Skewness of total_length: -0.29499010534783393
Skewness of head_length: -0.02320831402864604
Skewness of ear_size: 0.11130619639514261
Skewness of foot_length: 0.09481765086417197
Skewness of skull_width: 1.1244417448374424
Skewness of eye_diameter: 0.6337343026990503
Skewness of chest_circumference: -0.036450462205740136
Skewness of belly_circumference: 0.11758065235904504
```

After applying these approaches, most columns now have acceptable skewness, but skull_width remains highly skewed. To address this, the capping method was applied at the 95th percentile. Capping, or winsorization, replaces extreme values with a threshold (the 95th percentile in this case), reducing skewness while preserving the overall dataset (Bobbitt, 2021). For skull_width, this approach brings the distribution closer to normal, making it more suitable for analysis.

Before and after capped skull_width

	skull_width	skull_width_capped
count	99.000000	99.000000
mean	11.818358	11.643106
std	2.971781	2.482092
min	5.000000	5.000000
25%	10.000000	10.000000
50%	11.300000	11.300000
75%	13.050000	13.050000
max	23.600000	16.630000

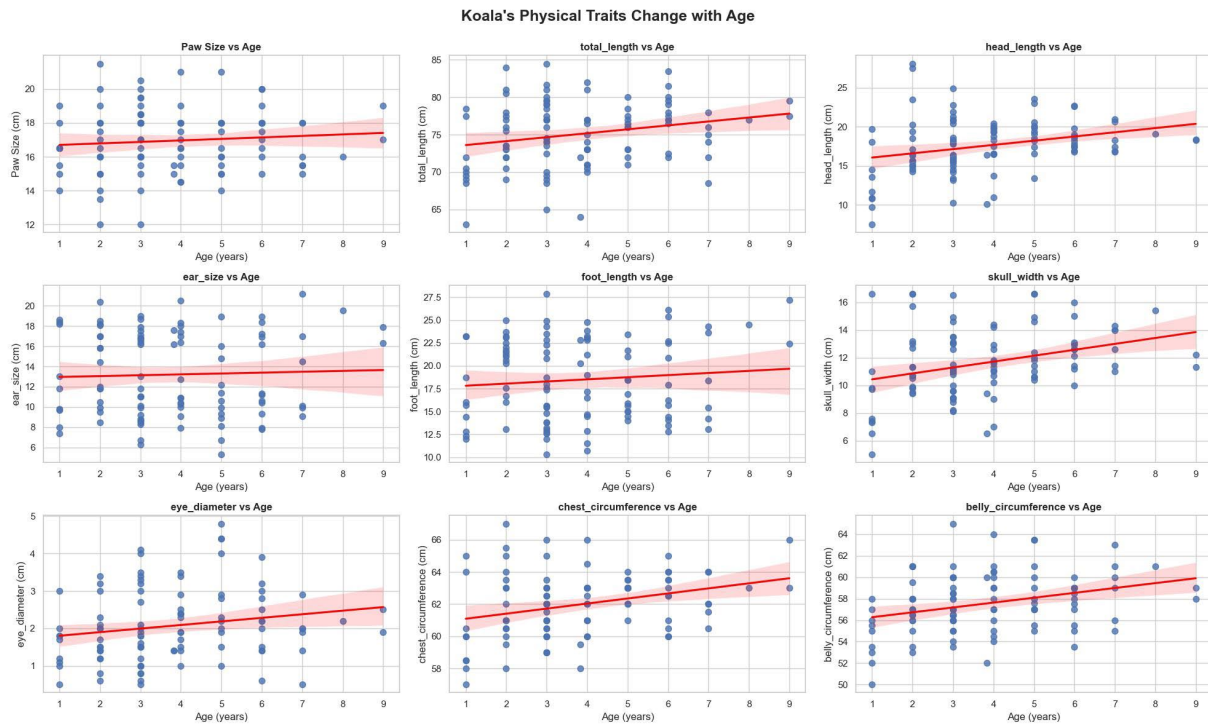
Skewness – after 2nd handling outlier

```
Skewness of age_in_years: 0.5623976380148998
Skewness of Paw Size: -0.014512497232162318
Skewness of total_length: -0.29499010534783393
Skewness of head_length: -0.02320831402864604
Skewness of ear_size: 0.11130619639514261
Skewness of foot_length: 0.09481765086417197
Skewness of skull_width: 0.03469736501697853
Skewness of eye_diameter: 0.6337343026990503
Skewness of chest_circumference: -0.036450462205740136
Skewness of belly_circumference: 0.11758065235904504
```

After capping the skull_width feature, the skewness is now within the acceptable range. A comparison of the statistics before and after capping shows that the adjustment did not compromise data integrity since the mean, median and interquartile range remain largely unchanged. As the cleansing process is now complete, the dataset is ready for further analysis.

Section 2: Visualisation

1) Overall Analysis of Koala Physical Traits Changes with Age



The set of scatter plots illustrates the changes in various physical characteristics of koalas as they age from 1 to 9 years. Each plot represents a different trait including paw size, total length, ear size, foot length, head length, skull width, eye diameter, chest circumference, and belly circumference, which tracked against the age of the koalas. Also, a red trend line in each graph helps visualize the overall trend, whether there is an increase, decrease, or stability in the measurements as koalas age. Below is an analysis regarding these trends.

Growth Trends:

- **Consistent Growth in Body Size:** Most of body measurements such as total length, chest circumference, and belly circumference show a consistent increase as koalas age. This growth trend might indicate good health and robustness, which are vital for their survival and reproductive success in the wild.
- **Head and Skull Development:** Both head length and skull width increase with age, which might be linked to the development of sensory organs and brain growth. This could influence behaviors such as foraging and social interactions, important aspects of koala ecology.

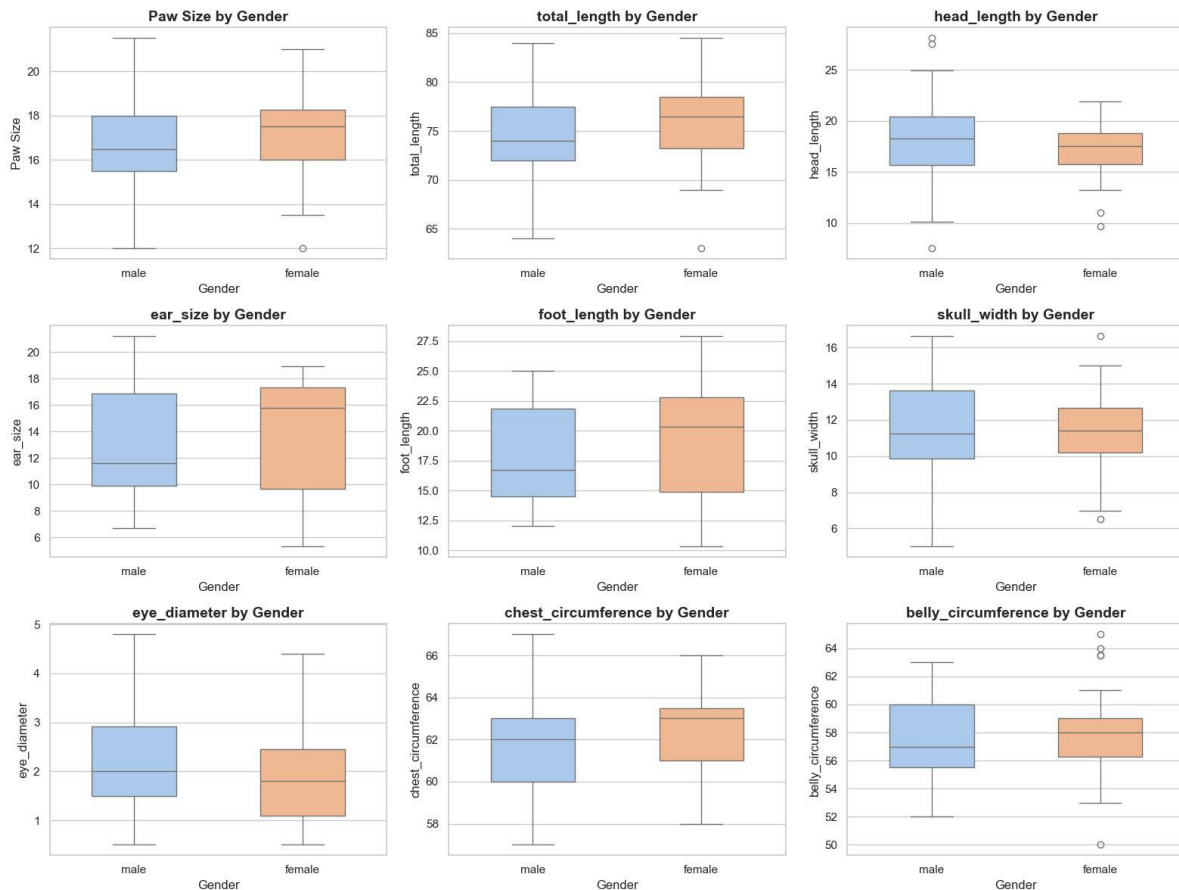
Stable Measurement:

- **Paw Size and Foot Length:** Interestingly, paw size and foot length exhibit relatively stable trends throughout the age span. This stability might suggest that these dimensions do not significantly impact koala's survival or that they reach their functional size early in life.
- **Ear Size and Eye Diameter:** Both features show slight increases but remain relatively consistent. The stability in ear size could be particularly significant, considering its role in thermoregulation and social signalling.

Individual Variability:

- **High Variance:** There is noticeable scatter in the data for all ages, indicating significant individual differences within the population. This might be influenced by genetics, health, and environmental conditions.

2) Koala Physical Traits by Gender



The box plots illustrate the distribution of various physical traits of koalas differentiated by gender. Each plot compares koala body measurements between male and female.

Differences in Body Size and Dimensions:

- **Total Length and Chest Circumference:** Male koalas typically exhibit a longer total body length and a larger chest circumference compared to females, indicating a larger overall body size.
- **Head Length and Skull Width:** Males also show longer head lengths and wider skulls, which might suggest a gender-related variation in skull morphology.
- **Belly Circumference:** Interestingly, females tend to have larger belly circumferences, which could be related to reproductive functions such as pouch and uterus accommodations.

Stable and Variant Measurements:

- **Paw Size and Foot Length:** Both genders have similar paw sizes and foot lengths, implying that these traits are less influenced by gender differences and more by other evolutionary or environmental factors.
- **Ear Size and Eye Diameter:** Females generally have larger ears and eye diameters. Larger ears in females might enhance their sensory capabilities for detecting predators or monitoring offspring, whereas larger eyes could be advantageous for nocturnal activities.

Variability:

- The plots show some variability and outliers, particularly in traits like skull width and belly circumference which differences are likely influenced by other genetic or environment factors.

Section 3: Analysis

1) Significant Difference in Mean Head Length Compared to 92.0 mm?

► State the Null Hypothesis and the Alternative Hypothesis

Null hypothesis (H_0): The mean head length of koalas is equal to 92.0 mm.

Alternative hypothesis (H_1): The mean head length of koalas is not equal to 92.0 mm

► Choose a significant level → a significant level of $\alpha = 0.05$

► Perform the One-Sample t-test

- Use the function `stats.ttest_1samp()` to perform a one-sample t-test to compares the sample mean of Koala head length to a value, 92 mm.
- Test Results: **T-statistic: -206.800** and **P-value: 3.1848e-131**

► Interpret the Results

- Since the p-value (3.1848e-131) is much smaller than the significance level of 0.05, **we reject the null hypothesis (H_0)**.
- **Conclusion:** There is overwhelming statistical evidence to suggest that the mean head length of koalas is **significantly different** from 92.0 mm. The negative t-statistic indicates that the sample mean is smaller than 92.0 mm.

2) Significant Difference in Mean Head Length between male and female koalas?

► State the Null Hypothesis and the Alternative Hypothesis

Null hypothesis (H_0): The mean head length of male koalas is equal to the mean head length of female koalas.

Alternative hypothesis (H_1): The mean head length of male koalas is not equal to the mean head length of female koalas (two-tailed test).

► Choose a significant level → a significant level of $\alpha = 0.05$

► Perform the two-Sample t-test

- Use the function `stats.ttest_1samp()` to perform a two sample t-test. This test compares the means of head length between male and female koalas.
- Test Results: **T-statistic: 1.0976** and **P-value: 0.2751**

► Interpret the Results

- The p-value is much larger than the chosen significance level of 0.05, we **fail to reject the null hypothesis (H_0)**.
- **Conclusion:** There is **no statistical evidence** to suggest that the mean head length of male koalas is significantly different from that of female koalas. Therefore, the head lengths of male and female koalas are likely to be similar based on this data.

3) Prediction of 'Total Length' of a Koala based on its 'Head Length'

► Set up and perform the OLS Regression Model

- Aiming to predict the total length of a koala based on its head length using a linear regression model.

Dependent variable: ***total_length***

Independent variable: ***head_length***

OLS formula: ***total_length ~ head_length***

- Using the `statsmodels` library to fit the OLS model with the data.

► OLS Regression Results (see Appendix A, Figure A-1)

- | | |
|---|--|
| • R-squared: 0.463 | Coefficients: |
| • Adj. R-squared: 0.458 | • Intercept: 60.6211 (<i>p-value < 0.000</i>) |
| • F-statistic: 83.76 (<i>p-value = 9.09e-15</i>) | • Head Length: 0.8229 (<i>p-value < 0.000</i>) |
| • Number of observations: 99 | |

► Model Interpretation

- **R-squared value of 0.463:** This suggests that approximately 46.3% of the variability in total length can be explained by head length.
- **Coefficient for head length (0.8229):** Indicates that for every 1 mm increase in head length, the total length increases by approximately 0.8229 mm.
- **Statistical Significance:** The p-value for the head length coefficient (less than 0.05) confirms that head length is a statistically significant predictor of total length.

► Assumption Checks

The model's results are valid, as the assumptions have been checked as following:

- **Linearity Check:** The total length is linearly related to head length.
- **Independent Observations:** Each observation of koala dataset is independent.
- **Normality Check:** A histogram of residuals (Figure A-2) with a kernel density estimate (KDE) shows a bell-shaped curve, suggesting normal distribution.
- **Homoscedasticity Check:** The residuals vs. fitted values plot (Figure A-3) shows a random scatter with no clear pattern, supporting the assumption of homoscedasticity.

► Conclusion

it is capable to predict the total length of a koala based on its head length. The linear regression analysis shows that head length is a statistically significant predictor of total length. However, the model only explains 46.3% of the variability in total length that can be explained by head length.

4) Prediction of 'Total Length' of a Koala based on its 'Head Length', 'Skull Width' and 'Foot Length'

► Set up and perform the OLS Multiple Linear Regression Model

- Aiming to predict the total length of a koala based on its head length, skull width and foot length, using a multiple linear regression model.

Dependent variable: **total_length**

Independent variable: **head_length, skull_width, foot_length**

- Adding the constant to independent variable for the intercept.
- Using the statsmodels library to fit the OLS model with the data.

► OLS Regression Results (see Appendix B, Figure B-1)

- | | |
|--|---|
| • R-squared: 0.511 | Coefficients: |
| • Adj. R-squared: 0.496 | • Intercept: 57.9552 (p-value < 0.000) |
| • F-statistic: 33.15 (p-value = 9.46e-15) | • Head Length: 0.5864 (p-value < 0.000) |
| • Number of observations: 99 | • Skull Width: 0.2590 (p-value = 0.200) |
| | • Foot Length: 0.2063 (p-value = 0.008) |

► Model Interpretation

- **R-squared value of 0.511:** This indicates that approximately 51.1% of the variability in total length can be explained by the head length, skull width, and foot length.
- **Head Length coefficient (0.5864):** For every 1 mm increase in head length, the total length increases by approximately 0.5864 mm.
- **Foot Length coefficient (0.2063):** For every 1 mm increase in foot length, the total length increases by approximately 0.2063 mm.
- **Skull Width:** Although the coefficient is positive, the p-value (0.200) suggests that it is not statistically significant in this model.

► Assumption Checks

The model's results are valid, as the regression analysis assumptions have been checked as following:

- **Normality Check:** The histogram of residuals (Figure B-2) and the QQ plot (Figure B-3) indicate that the residuals are approximately normally distributed.
- **Homoscedasticity Check:** The residuals vs. fitted values (Figure B-4) plot shows a random scatter, supporting the assumption of homoscedasticity.
- **Multicollinearity Check:** The Variance Inflation Factor (VIF) was calculated to detect multicollinearity, which occurs when independent variables are highly correlated, leading to overlapping information. The VIF scores indicate that multicollinearity is within acceptable limits, ensuring the reliability of the model's coefficients.
 - Head Length VIF: 2.712 (Moderate multicollinearity)
 - Skull Width VIF: 2.581 (Moderate multicollinearity)
 - Foot Length VIF: 1.165 (Low multicollinearity)
- **Linearity Check:** Scatter plots between total length and each predictor variable (head length, skull width, and foot length) (Figure B-5) show a positive linear relationship. This confirms that the linearity assumption is met.

► Conclusion

The multiple linear regression model indicates that head length and foot length are significant predictors of total length in koalas, while skull width does not appear to have a statistically significant impact. The model explains 51.1% of the variation in total length, suggesting that while head length and foot length are important factors, there are likely other variables influencing total length that have not been included in the model. Including more predictors could provide a comprehensive analysis.

5) Examining the Impact of Habitat (State) on Koala Physical Characteristics

► One-Hot Encoding the Categorical Variable 'Habitat'

Since 'habitat' (or state) is a categorical variable, it needed to be transformed into a numerical format. One-hot encoding was applied, converting the 'habitat' variable into binary columns to represent different states (VIC and QLD).

► Selecting Physical Characteristics for Correlation Analysis

After encoding, I have selected all of physical characteristics along with the encoded 'habitat' columns ('habitat_VIC'). Then, computed the correlation matrix between the physical characteristics and the habitat columns using 'corr()'. This helped identify whether certain physical traits are positively or negatively correlated with a particular habitat.

```
characteristic_cols = ['Paw Size', 'total_length', 'head_length', 'ear_size',
                      'foot_length', 'skull_width', 'eye_diameter',
                      'chest_circumference', 'belly_circumference']
```

► Findings from Correlation

- **Ear Size (0.878375) and Foot Length (0.805319):** Strong positive correlations indicate that koalas in VIC tend to have larger ear sizes and longer foot lengths compared to QLD koalas.

- **Paw Size (-0.477885):** A moderate negative correlation shows that koalas in VIC tend to have smaller paw sizes than those in QLD.
- **Total Length (0.085825) and Chest Circumference (0.230076):** These traits show small positive correlations with VIC, meaning koalas in VIC may have slightly longer bodies and larger chests compared to those in QLD. However, these differences are not as pronounced as ear size and foot length.
- **Skull Width (-0.057307) and Eye Diameter (-0.217716):** These traits exhibit weak negative correlations, implying minimal differences between koalas in VIC and QLD in terms of skull width and eye diameter.
- Also, there is a boxplot analysis that demonstrates the distribution of various koala physical characteristics, differentiated by habitat or state (see Appendix C).

	index	habitat VIC
0	ear_size	0.878375
1	foot_length	0.805319
2	chest_circumference	0.230076
3	total_length	0.085825
4	belly_circumference	0.032651
5	head_length	0.003633
6	skull_width	-0.057307
7	eye_diameter	-0.217716
8	Paw Size	-0.477885

► Conclusion

The analysis suggests that habitat (state) does have an impact on certain koala physical characteristics.

- Koalas in VIC tend to have larger ear sizes and longer foot lengths compared to those in QLD.
- QLD koalas tend to have larger paws than VIC koalas.
- Other traits; total length and chest circumference, show only slight differences between states.

Environmental factors in VIC and QLD appear to influence the physical characteristics of koalas, though findings might vary. In VIC, where the climate is cooler and vegetation denser, koalas tend to have larger ears, potentially aiding in both thermoregulation and better hearing in dense, low-visibility areas. Longer feet may also assist in navigating thicker underbrush. Meanwhile, in the warmer climate of QLD, koalas typically have larger paws and shorter feet, which could improve grip when climbing thinner, smooth-barked trees. These adaptations reflect how regional conditions influence koala morphology (Australian Koala Foundation, n.d.).

6) Correlated Factors with the total length of Koala

► Prepare data for correlation analysis'

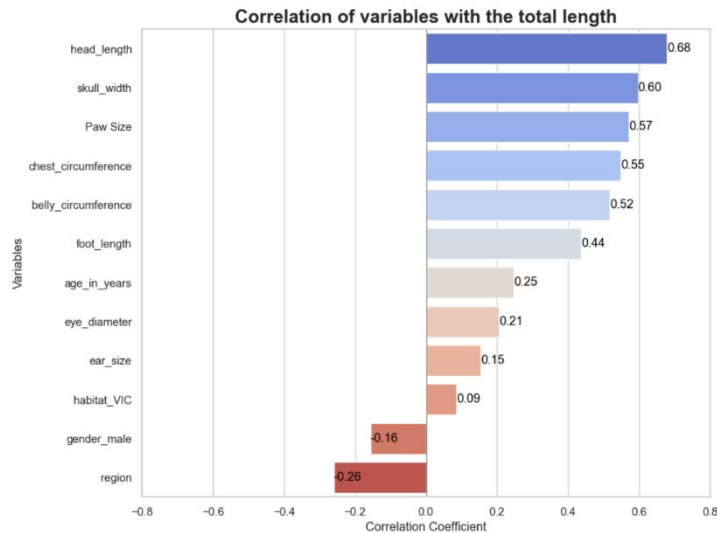
Using the encoded dataset from previous which categorical variables ('habitat' and 'gender') already converted to numeric to allow a proper correlation. Also, the 'koala_id' is dropped as it is an identifier and irrelevant for correlation purposes.

► Extract and Sort Correlations corresponding with Total Length

Isolates the correlation values that directly relate to the 'total_length' variable, as we are specifically interested in the factors that correlate with total body length. Then, sorted the correlation values in descending order to highlight from the highest to the least.

► Visualize the correlations

I have created a bar plot to visualize the correlation coefficients, making it easier to interpret which factors have the strongest and weakest relationships with total length.



► Key Finding

Head length (0.68): The strongest positive correlation with total length. This suggests that as the head length increases, the total length of the koala also tends to increase.

Skull Width (0.60): Another strong positive correlation, indicating that koalas with wider skulls also tend to have longer bodies.

Paw Size (0.57): Koalas with larger paw sizes tend to have longer total body lengths.

Chest (0.55)/ Belly Circumference (0.52): These also show strong positive correlations with total length, suggesting that koalas with larger circumferences tend to be longer in body size.

Foot Length (0.44): A moderate positive correlation, indicating that longer foot lengths are somewhat associated with longer body lengths.

Weaker Correlations:

- **Age in Years (0.25), Eye Diameter (0.21), and Ear Size (0.15)** show weaker positive correlations.
- **Habitat_VIC (0.09):** This small positive correlation suggests that habitat (state) may have a minimal effect on the koala's total length whereas VIC may have slightly longer in body length than QLD.
- **Gender_male (-0.16) and Region (-0.26):** Negative correlations indicate that male koalas and koalas from certain regions may have slightly shorter total lengths.

► Conclusion

The analysis shows that head length, skull width, and paw size are the most strongly correlated factors with the total length of a koala in which seems reasonable, as larger head and paw sizes are often associated with overall body size. Traits such as chest circumference and belly circumference also play important roles. Weaker correlations were found with age, eye diameter, and ear size, while habitat and gender showed minimal or negative correlations with total length.

Section 4: Recommendation

Aligning with my findings, I have identified the effective koala conservation as following approaches:

- **Habitat preservation tailored to regional traits:** As koalas in different areas show distinct adaptations. For example, VIC koalas have larger ears and foot lengths, while QLD koalas have larger paws. VIC conservation efforts should focus on cooler, denser eucalyptus forests that support traits like longer foot lengths and small stronger paws, while QLD koalas benefit from open habitats with sufficient climbing trees. **Tailored habitat restoration** strategies must address the specific environmental and biological conditions of koalas in each region, as recommended by the Spot Assessment Technique, which emphasizes localized conservation strategies to ensure effective habitat protection (Phillips & Callaghan, 2011).
- **Growth Tracking Program:** As findings indicate that head length, skull width, paw size has strong correlation with total length which reflects overall size and physical development. They could serve as useful indicators of proper growth and development. Regular monitoring of these traits, particularly in wild populations and rehabilitation centers, can help conservationists track whether koalas are growing as expected, allowing them to flag potential health issues if significant deviations from normal growth patterns are observed.

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Appendix A: Linear Regression Model Results and Evaluation

This appendix contains the results from the linear regression analysis conducted to predict total koala length based on head length. It includes the OLS model summary, a histogram of residuals, and Residuals vs. Fitted Values plots for model evaluation.

OLS Regression Results						
Dep. Variable:	total_length		R-squared:	0.463		
Model:	OLS		Adj. R-squared:	0.458		
Method:	Least Squares		F-statistic:	83.76		
Date:	Mon, 14 Oct 2024		Prob (F-statistic):	9.09e-15		
Time:	11:02:00		Log-Likelihood:	-254.19		
No. Observations:	99		AIC:	512.4		
Df Residuals:	97		BIC:	517.6		
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	60.6211	1.614	37.549	0.000	57.417	63.825
head_length	0.8229	0.090	9.152	0.000	0.644	1.001
Omnibus:	0.496	Durbin-Watson:	1.640			
Prob(Omnibus):	0.781	Jarque-Bera (JB):	0.640			
Skew:	-0.070	Prob(JB):	0.726			
Kurtosis:	2.632	Cond. No.	90.8			

Figure A-1: OLS Regression Results

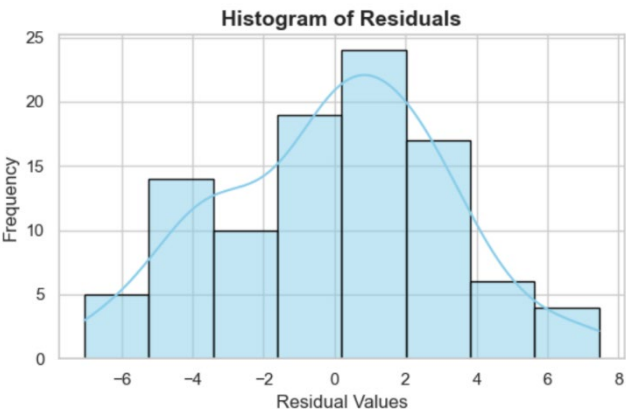


Figure A-2: Histogram of Residuals

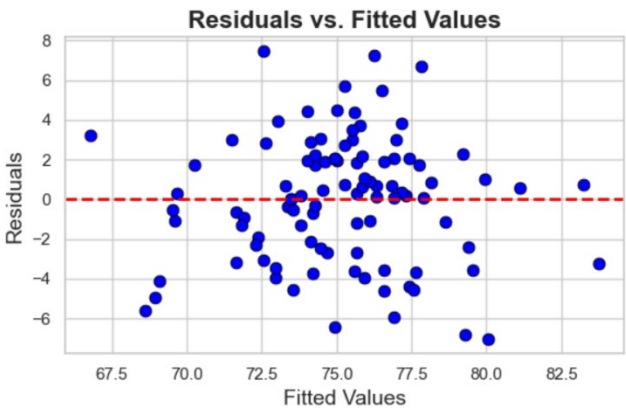


Figure A-3: Residuals vs. Fitted Values

Figure A-2: The histogram of residuals shows they are approximately normally distributed. Furthermore, Figure A-3: the Residuals vs. Fitted Values plot displays no discernible pattern, indicating homoscedasticity, or constant variance of residuals across fitted values. These findings confirm that the model meets the key regression assumptions, supporting the validity of the analysis.

Appendix B: Multiple Linear Regression Model Results and Evaluation

This appendix contains the results from the multiple linear regression analysis conducted to predict total koala length based on head length, skull width and foot length. It includes the OLS regression summary, a histogram of residuals, and Residuals vs. Fitted Values plots for model evaluation.

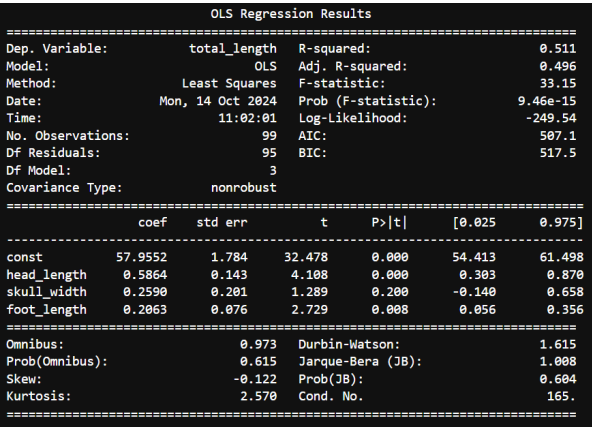


Figure B-1: OLS Regression Result

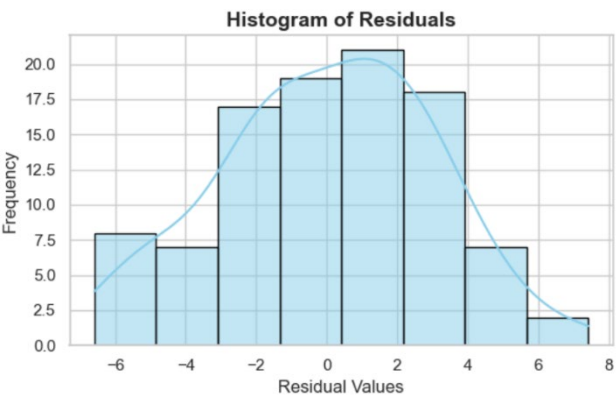


Figure B-2: Histogram of Residuals

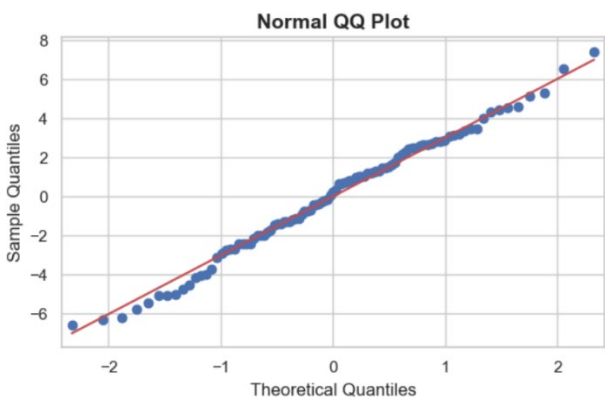


Figure B-3: Normal QQ plot

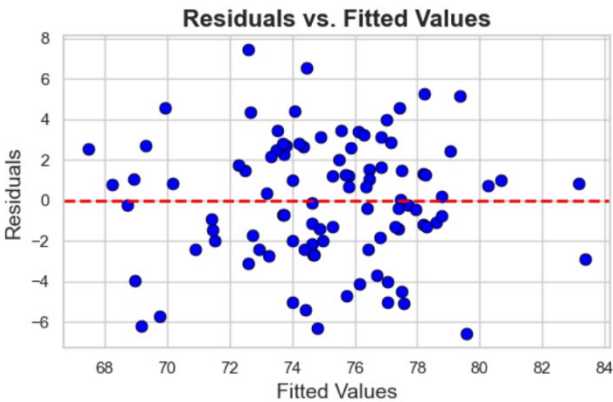


Figure B-4: Residuals vs. Fitted Values

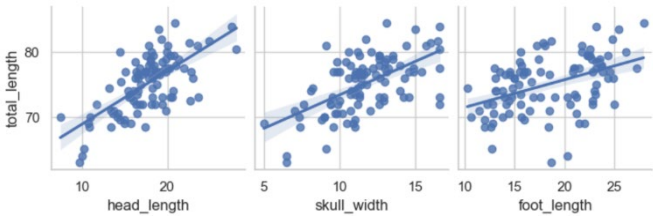


Figure B-5: Linear relationship between total length and the predictors

The histogram of residuals (Figure B-2) and the normal Q-Q plot (Figure B-3) confirm that the residuals are approximately normally distributed. Additionally, the Residuals vs. Fitted Values plot (Figure B-4) reveals no discernible pattern, indicating homoscedasticity, where the residuals maintain constant variance across the range of fitted values. Furthermore, Figure B-5 demonstrates a linear relationship between total length and its predictors. These findings suggest that the model meets the key assumptions of regression, supporting the validity of the analysis.

Appendix C: Boxplot of Koala Physical Characteristics by Habitat

This appendix presents boxplots illustrating the distribution of various koala physical characteristics, differentiated by habitat or state. The boxplots allow for a comparative analysis of key traits such as ear size, foot length, and head length, highlighting the variations between koalas from different regions, such as Victoria (VIC) and Queensland (QLD).

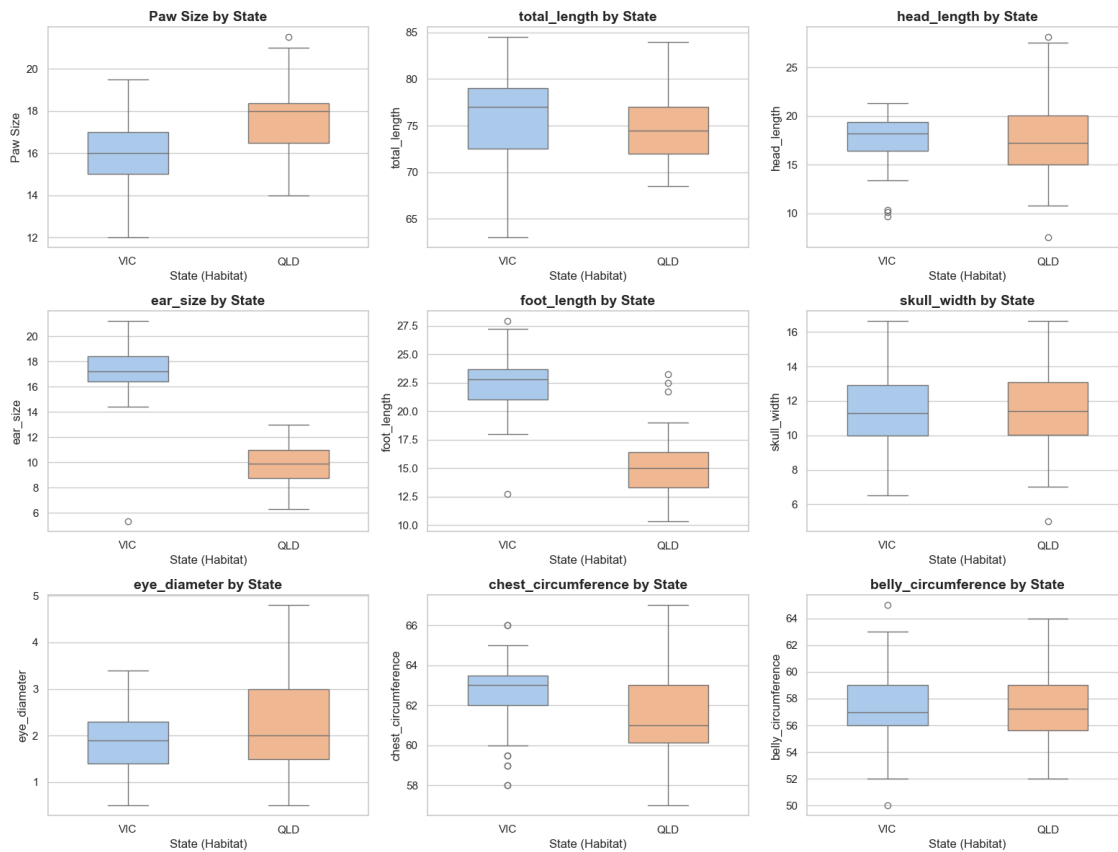


Figure C-1: Boxplots of Koala Physical Characteristics by Habitat/State

Regarding to the boxplot, the insights can be gathered as follow:

- Total Length: Koalas in Queensland (QLD) tend to have a longer total body length compared to those in Victoria (VIC).
- Paw Size: There's a slight variation in paw size, with QLD koalas showing slightly larger paws.
- Head Length: QLD koalas also appear to have longer heads on average than those in VIC.
- Ear Size: Ear sizes of koalas are notably smaller in QLD compared to VIC.
- Foot Length: Foot length is quite similar between the states, though QLD shows a slightly higher average.
- Skull Width: QLD koalas have wider skulls compared to those in VIC.
- Eye Diameter: VIC koalas have larger eye diameters than those in QLD.
- Chest Circumference: Chest circumference is greater in QLD koalas.
- Belly Circumference: Belly circumference is relatively similar between the states, with a slight tendency for larger sizes in QLD.